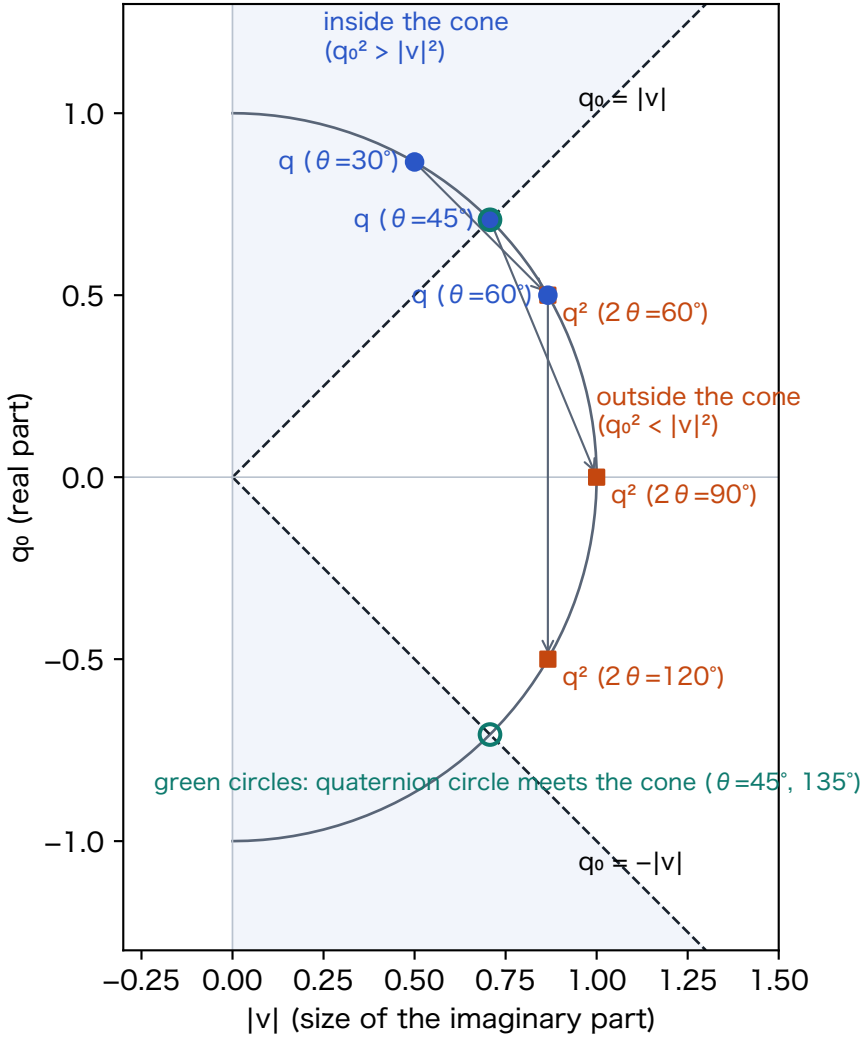Figure 4. Light cone and the true square: $q^2 = \cos 2\theta + n \sin 2\theta$, real part $q_0^2 - |v|^2 = \cos 2\theta$. At $\theta = 45°$, q lies on the cone and q^2 comes to real part 0 (the equator)